

Exhaustive Hunter searches for quintic A_5 fields of prescribed prime-power discriminant, with an application to icosahedral ramification at 1951 and 15733

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Abstract

Let $K/\mathbb{Q}$ be a quintic field with $\text{Gal}(\tilde{K}/\mathbb{Q}) \cong A_5$. We carry out exhaustive Hunter searches, using the sharp Hermite constant $\gamma_4 = \sqrt{2}$, over the complete Minkowski box attached to two prescribed discriminants. First, we prove computationally that *no* quintic A_5 field of any signature has $d_K = 1951^2$; the search covers 6.056×10^{10} coefficient vectors and no candidate survives verification of the field discriminant. Second, at $d_K = 15733^2$ we exhibit three monogenic totally real quintic A_5 fields ramified only at 15733, including

$$x^5 - x^4 - 18x^3 - 7x^2 + 5x + 1, \quad d_K = 15733^2 = 247\,527\,289.$$

We record the elementary group-theoretic constraints on tame inertia in A_5 and $\text{SL}(2, 5)$ that make these two discriminants the relevant test cases for even icosahedral Artin representations of prime conductor, and we note the consequence: an even icosahedral representation unramified outside 1951 must have inertia of order 5 at 1951, hence $d_K = 1951^4$ and Artin conductor 1951^2 . We also give a verified PARI/GP implementation of the classical passage from an A_5 sextic to its quintic resolvent by synthemetic totals, together with two failure modes of that construction that produce plausible but incorrect output.

Contents

1	Introduction	2
1.1	Motivation	2
1.2	Results	2
1.3	What is and is not proved	2
2	Hunter’s theorem and the search box	3
3	Tame inertia in A_5 and $\text{SL}(2, 5)$	4
4	The search	5
4.1	The predicate	5
4.2	Algorithm	5
4.3	Cost model and evaluation	5
4.4	Controls	6

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5 Results at $d = 1951^2$	6
6 Results at $d = 15733^2$	6
7 From an A_5 sextic to its quintic resolvent	7
8 Consequences for icosahedral ramification at 1951	8
9 Reproducibility	9

1 Introduction

1.1 Motivation

Let $\rho: G_{\mathbb{Q}} \rightarrow \mathrm{GL}_2(\mathbb{C})$ be an irreducible Artin representation whose projective image is A_5 . Such ρ are called *icosahedral*, and are called *even* when $\det \rho(c) = +1$ for complex conjugation c . The odd case is a theorem of Khare–Wintenberger; the even case remains open and is expected to correspond to Maass forms of eigenvalue $1/4$.

The concrete objects attached to such a ρ are number fields: the degree 60 projective field $\tilde{K}^{\mathrm{proj}}$, its quintic and sextic resolvents K_5 and K_6 , and (when ρ has trivial determinant) a degree 24 field with group $\mathrm{SL}(2, 5) = 2.A_5$. Numerical work on even icosahedral L -functions therefore begins by locating a quintic A_5 field of prescribed ramification. This paper reports two such determinations, both carried out to exhaustion rather than by heuristic search.

1.2 Results

Throughout, d_K denotes the discriminant of the field, and $\mathrm{disc}(f)$ the discriminant of a defining polynomial; the two differ by the square of the index $[\mathcal{O}_K : \mathbb{Z}[\theta]]$.

Theorem 1.1. *No quintic field K with $\mathrm{Gal}(\tilde{K}/\mathbb{Q}) \cong A_5$, of any signature, has $d_K = 1951^2 = 3\,806\,401$.*

Theorem 1.2. *There exist totally real quintic fields K with $\mathrm{Gal}(\tilde{K}/\mathbb{Q}) \cong A_5$, $d_K = 15733^2 = 247\,527\,289$, and $[\mathcal{O}_K : \mathbb{Z}[\theta]] = 1$. Three defining polynomials are*

$$x^5 - x^4 - 18x^3 - 7x^2 + 5x + 1, \quad x^5 - 17x^3 - 23x^2 - x + 5, \quad x^5 - 17x^3 + 23x^2 - x - 5.$$

Each is irreducible with five real roots, has Galois group A_5 , and nfdisc equal to 15733^2 exactly, so 15733 is the only ramified prime.

Theorem 1.1 is a negative result obtained by exhausting the complete Hunter box; Theorem 1.2 is a positive result obtained from 0.4% of the corresponding box, since a single hit suffices for existence. Section 8 explains why these two primes are the relevant ones, and Section 7 gives the sextic-to-quintic construction used to move between resolvents.

1.3 What is and is not proved

We are explicit about the epistemic status of each statement.

- Theorems 1.1 and 1.2 are computational, but their logical form differs. Theorem 1.2 is constructive and is verified independently in PARI/GP by `nfdisc`, `polgalois` and `polsturm`; it does not depend on the exhaustiveness of any search. Theorem 1.1 depends on exhausting a finite box, and hence on the correctness of the enumeration; we describe the controls in Section 4.4.
- The claim in Section 8 that 1951 must carry inertia of order 5 depends additionally on the hypothesis that the field is unramified outside 1951, and on the relation between the Artin conductor and d_K recorded in Lemma 3.2. We do not prove that any even icosahedral representation of conductor 1951^2 exists.
- We make no novelty claim for the fields of Theorem 1.2. A totally real A_5 quintic of discriminant 15733^2 may already appear in existing tables; our contribution is the exhaustive framework and the verification, not necessarily the field.
- The completeness of published enumerations of totally real quintic fields below 2×10^7 is used only for context in Section 8 and is not used in the proof of any theorem above.

2 Hunter's theorem and the search box

We use the following form of Hunter's theorem; see Cohen [1, Thm. 6.4.2].

Theorem 2.1 (Hunter). *Let K be a number field of degree n with discriminant d_K . There exists $\theta \in \mathcal{O}_K \setminus \mathbb{Z}$ such that, writing $a_1 = \text{Tr}(\theta)$ and $T_2(\theta) = \sum_{i=1}^n |\theta_i|^2$ for the sum over all complex embeddings,*

$$0 \leq a_1 \leq \frac{n}{2}, \quad T_2(\theta) \leq \frac{a_1^2}{n} + \gamma_{n-1} \left(\frac{|d_K|}{n} \right)^{1/(n-1)},$$

where γ_m denotes the Hermite constant.

For $n = 5$ the relevant Hermite constant is $\gamma_4 = \sqrt{2}$. Writing $B(a_1, d) = a_1^2/5 + \sqrt{2}(d/5)^{1/4}$ we obtain, for our two targets,

$$B(\cdot, 1951^2) \leq 42.57, \quad B(\cdot, 15733^2) \leq 118.63.$$

Remark 2.2. Using the crude bound $\gamma_{n-1} \leq n-1$ in place of $\gamma_4 = \sqrt{2}$ inflates the first bound from 42.57 to 118.2. Since the cost of the search grows like B^7 (Section 4.3), the sharp constant is worth a factor of roughly 10^3 .

Write $f(x) = x^5 - e_1x^4 + e_2x^3 - e_3x^2 + e_4x - e_5$, so that e_k is the k -th elementary symmetric function of the roots. Put $p_2 = \sum_i \theta_i^2 = e_1^2 - 2e_2$. Then $a_1 = e_1 \in \{0, 1, 2\}$, and e_2 is determined by (a_1, p_2) .

Lemma 2.3. *If $T_2(\theta) \leq B$ then $|p_2| \leq B$, and for $1 \leq k \leq 5$,*

$$|e_k| \leq \binom{5}{k} \left(\frac{B}{k} \right)^{k/2}.$$

Proof. Each e_k is a sum of $\binom{5}{k}$ products of k roots, and by AM-GM $|\theta_{i_1} \cdots \theta_{i_k}| \leq (|\theta_{i_1}|^2 + \cdots + |\theta_{i_k}|^2)/k)^{k/2} \leq (T_2/k)^{k/2}$. For p_2 , note $|p_2| = |\sum \theta_i^2| \leq \sum |\theta_i|^2 = T_2$. $\square$

Lemma 2.3 is valid for every signature. This matters: for a totally real field $p_2 = T_2$ and the box shrinks with p_2 , but in general p_2 is a power sum and only $|p_2| \leq T_2$ holds, so p_2 ranges over $[-B, B]$ and every box must be sized by B .

Lemma 2.4 (Newton, real-rooted case). *If f has all roots real, then with $c_k = e_k/\binom{5}{k}$ one has $c_k^2 \geq c_{k-1}c_{k+1}$ for $1 \leq k \leq 4$.*

These inequalities prune the totally real search but are invalid in general; they are not used for Theorem 1.1.

3 Tame inertia in A_5 and $\mathrm{SL}(2, 5)$

Let $L/\mathbb{Q}$ be Galois with group $G \subseteq A_5$ and let $p \nmid |A_5| = 60$ be ramified. Then inertia $I \subseteq G$ at p is cyclic of order $n \in \{2, 3, 5\}$, and ramification is tame.

Lemma 3.1. *Let σ generate I and let τ be a Frobenius lift, so $\tau\sigma\tau^{-1} = \sigma^p$. Then the automorphism $\sigma \mapsto \sigma^p$ of I lies in the image of $N_G(I) \rightarrow \mathrm{Aut}(I)$. Consequently, for $G = A_5$:*

1. $n = 2$: $\mathrm{Aut}(\mathbb{Z}/2)$ is trivial and no condition arises.
2. $n = 3$: $N_{A_5}(I)/C_{A_5}(I) \cong \mathbb{Z}/2$ realises both automorphisms, and $p \equiv \pm 1 \pmod{3}$ holds for all $p \neq 3$.
3. $n = 5$: $N_{A_5}(I) \cong D_5$ and $C_{A_5}(I) = I$, so $N/C \cong \mathbb{Z}/2$ and necessarily

$$p \equiv \pm 1 \pmod{5}.$$

Lemma 3.2. *Let I be tame cyclic of order n acting on a transitive G -set X . Then $v_p(\mathrm{disc})$ of the corresponding field equals $|X| - \#\{I\text{-orbits on } X\}$. For A_5 on 5 and 6 points, and for $\mathrm{SL}(2, 5)$ on the 24 cosets of a Sylow 5-subgroup, the values are as in Table 1.*

$ I $	type on 5	$v_p(d_5)$	type on 6	$v_p(d_6)$	$v_p(\mathrm{disc}_{24})$
2	(2, 2, 1)	2	(2, 2, 1, 1)	2	18
3	(3, 1, 1)	2	(3, 3)	4	16 or 20
5	(5)	4	(5, 1)	4	16 or 20

Table 1: Tame inertia of order n at $p \nmid 60$: cycle types and discriminant exponents. In degree 24 the two values correspond to the two possible orders of the preimage of I in $\mathrm{SL}(2, 5)$ (3 or 6; 5 or 10). The preimage of an involution is cyclic of order 4, giving 4^6 and exponent 18.

Corollary 3.3. *If $\mathrm{Gal}(\tilde{K}/\mathbb{Q}) \cong A_5$ then d_K is a perfect square, and if moreover K is ramified only at a single prime $p \nmid 60$ then $d_K = p^2$ or $d_K = p^4$, the latter only if $p \equiv \pm 1 \pmod{5}$.*

Lemma 3.4. *If ρ is even icosahedral then the quintic resolvent K_5 is totally real.*

Proof. Evenness gives $\rho(c) = \pm I$ when $\det \rho$ is trivial. The element $-I$ is central in $\mathrm{SL}(2, 5)$, so in either case the image of c in $A_5 = \mathrm{PSL}(2, 5)$ is trivial. Hence c acts trivially on the five points and K_5 has five real embeddings. $\square$

4 The search

4.1 The predicate

For a monic integral f with root θ generating K ,

$$\text{disc}(f) = d_K \cdot [\mathcal{O}_K : \mathbb{Z}[\theta]]^2.$$

Hunter's theorem supplies a generator of small T_2 but says nothing about the index. Hence the correct test for a target d is

$$\text{disc}(f) > 0, \quad d \mid \text{disc}(f), \quad \text{disc}(f)/d \text{ a perfect square}, \quad (1)$$

and *not* $\text{disc}(f) = d$. Condition (1) is necessary but far from sufficient; every surviving candidate must be settled by computing d_K itself.

Remark 4.1. Condition (1) admits two distinct failure modes, both observed in Section 5. A candidate may satisfy it with $d \mid d_K$ but $d_K \neq d$, the field being ramified at additional primes; or it may satisfy it with $d \nmid d_K$ altogether, the prime dividing the index rather than the discriminant. Reporting raw hits without computing d_K therefore produces false positives of two different kinds.

4.2 Algorithm

Algorithm 4.2. Given a target d and a signature mode:

1. For $a_1 \in \{0, 1, 2\}$ set $B = a_1^2/5 + \sqrt{2}(d/5)^{1/4}$.
2. For each integer p_2 with $p_2 \equiv a_1^2 \pmod{2}$ and $|p_2| \leq B$ (totally real mode: $1 \leq p_2 \leq B$), set $e_2 = (a_1^2 - p_2)/2$.
3. Let $S = p_2$ in totally real mode and $S = B$ otherwise, and enumerate $|e_k| \leq \binom{5}{k}(S/k)^{k/2}$ for $k = 3, 4, 5$ by Lemma 2.3.
4. In totally real mode, discard (e_3, e_4, e_5) violating Lemma 2.4.
5. Test (1).
6. For each survivor, test irreducibility and the presence of a Frobenius cycle type $(3, 1, 1)$, which together with disc square certifies A_5 .
7. Emit survivors for verification of d_K by `nfdisc`.

4.3 Cost model and evaluation

With b_k the bound of Lemma 2.3, the work in a cell is proportional to $b_3 b_4 b_5 \asymp p_2^6$ in totally real mode, so the total is $\asymp \int^B p^6 dp \asymp B^7$. Empirically

$$\#\{\text{triples}\} \approx 0.0092 B^7,$$

which reproduces the measured 2.74×10^9 at $B = 42.57$ to within 15%.

The inner loop evaluates $\text{disc}(f)$ on a lattice. Expanded, the quintic discriminant has 59 monomials in $(e_1, \dots, e_5)$; but it is a *quartic* in e_5 ,

$$\text{disc}(f) = A_4(e_1, \dots, e_4) e_5^4 + A_3 e_5^3 + A_2 e_5^2 + A_1 e_5 + A_0,$$

so evaluating the A_i once per (e_3, e_4) and applying Horner in e_5 replaces 59 monomial evaluations by four fused multiply-adds per lattice point. Measured on one core this raised throughput from 7.9×10^5 to 3.5×10^7 points per second, a factor of 44; the gain exceeds the arithmetic-operation ratio because the expanded form allocates many large intermediate arrays and saturates memory bandwidth under multiprocessing.

All arithmetic is exact in 64-bit integers. Over the boxes used here $|\text{disc}(f)| < 1.9 \times 10^{18} < 2^{63} - 1$, which we verified at the box corners.

4.4 Controls

The enumeration was validated against targets whose answer is known independently.

- $d = 1762^2 = 3\,104\,644$. The search returns ten defining polynomials, all totally real with $\text{Gal} \cong A_5$; `nfdisc` confirms $d_K = 2^2 \cdot 881^2 = 3\,104\,644$ for all ten, with indices 4, 6, 6, 9, 9, 16, 24, 24, 54, 111. The minimal index 4 agrees with the published value for this field.
- $d = 2277^2 = 5\,184\,729$. The search returns a monogenic generator, $\text{disc}(f)/d = 1^2$, at $p_2 = 18$.
- *Truncation control*. Restricting to $|p_2| \leq 0.45 B$, which is 0.4% of the box, still recovers the monogenic generator at $d = 2277^2$ in one second.

5 Results at $d = 1951^2$

Two independent passes were made.

- *Totally real*, using Lemma 2.4: 2 741 460 250 coefficient triples, no surviving field.
- *All signatures*, no Newton pruning, p_2 ranging over $[-42, 42]$: 60 561 616 546 triples. This pass strictly contains the first, since its boxes are sized by $B \geq p_2$ and no pruning is applied.

The second pass produced 19 raw hits satisfying (1), comprising 14 distinct irreducible A_5 polynomials, all of signature $(1, 2)$. Table 2 gives their field discriminants. Not one equals 1951^2 , which proves Theorem 1.1.

Remark 5.1. Entry 12 has $d_K = 2^4 \cdot 13^2 \cdot 31^2 = 2\,598\,544$, an A_5 quintic of signature $(1, 2)$ whose discriminant lies below $1762^2 = 3\,104\,644$. This is consistent with published tables, which enumerate the totally real case.

6 Results at $d = 15733^2$

Here $15733 \equiv 3 \pmod{5}$, so by Lemma 3.1 inertia of order 5 is impossible and, by Corollary 3.3, a field ramified only at 15733 must have $d_K = 15733^2$ exactly. Unlike the case of 1951, there is no alternative exponent, so this target admits a decisive positive answer.

The calibration in Section 4.4 showed that low-index generators occur at low p_2 : the minimal- p_2 hit had index 6 at $0.554 B$ for $d = 1762^2$ and index 1 at $0.399 B$ for $d = 2277^2$. Accordingly the search was run truncated to $|p_2| \leq 0.45 B$, costing $0.45^7 \approx 0.4\%$ of the full box, and terminated in 258 seconds on a single core after 1.52×10^{10} triples.

#	d_K	index	1951 ramified?
1	$5^2 \cdot 37^2 \cdot 1951^2$	3	yes
2	$11^2 \cdot 1951^2$	11	yes
3	$53^2 \cdot 479^2$	1951	no
4	$3^4 \cdot 41^2 \cdot 1951^2$	1	yes
5	$2^4 \cdot 5^2 \cdot 19^2 \cdot 1951^2$	2	yes
6	$3^2 \cdot 17^2 \cdot 1951^2$	27	yes
7	$53^2 \cdot 1951^2$	1	yes
8	$3^4 \cdot 5^2 \cdot 1951^2$	9	yes
9	$3^4 \cdot 5^2 \cdot 13^2 \cdot 1951^2$	1	yes
10	42467^2	1951	no
11	$2^2 \cdot 23^2 \cdot 1951^2$	16	yes
12	$2^4 \cdot 13^2 \cdot 31^2$	1951	no
13	$5^2 \cdot 11^2 \cdot 1951^2$	22	yes
14	$2^2 \cdot 19^2 \cdot 1951^2$	1	yes

Table 2: The 14 distinct A_5 candidates at $d = 1951^2$ surviving predicate (1), after computing d_K . Eleven carry 1951^2 but are ramified elsewhere; three have 1951 dividing the index only, with 1951 unramified. Both failure modes of Remark 4.1 occur.

Three hits occurred, all with $\text{disc}(f)/d = 1^2$ and $p_2 \in \{34, 34, 37\}$, that is at 0.287–0.312 of the Hunter bound. PARI/GP verification gives, for each of the three polynomials of Theorem 1.2,

$$\text{nfdisc} = 247\,527\,289 = 15733^2, \quad \text{index} = 1, \quad \text{polsturm} = 5, \quad \text{polgalois} = [60, 1, 1, "A_5"].$$

Since the factorisation of d_K consists of the single prime 15733 to the exponent 2, no other prime ramifies. This proves Theorem 1.2.

Remark 6.1. A truncated search is decisive only in the positive direction. Had it returned nothing, no conclusion would follow. The negative result of Section 5 correspondingly uses the complete box.

7 From an A_5 sextic to its quintic resolvent

Both resolvents of an A_5 extension are useful, and passing between them avoids constructing the degree 60 splitting field, for which `galoisinit` is often impractical.

The group $A_5 \cong \text{PSL}(2, 5)$ acts on the six points of $\mathbb{P}^1(\mathbb{F}_5)$ with point stabiliser D_5 . Among the 15 perfect matchings (*synthemes*) of the six points, this action has orbits of sizes 5 and 10; the orbit of size 5 is a *synthematic total*, and the stabiliser of one of its synthemes is A_4 . The induced action on the five synthemes of the total is the 5-point action of A_5 . Hence, for a syntheme $\{ab \mid cd \mid ef\}$ and roots $r_1, \dots, r_6$ of the sextic, the quantity

$$t = (r_a + r_b)(r_c + r_d)(r_e + r_f) \tag{2}$$

takes five values that are permuted by Gal as A_5 acts on five points, so they are the roots of the quintic resolvent.

There are six synthematic totals on six letters and Gal preserves exactly one of them. Since numerical root-finders return roots in an arbitrary order, the correct total must be determined rather than assumed. It is detected unambiguously by rationality of the resulting elementary symmetric functions.

source	stable total	$\max_k [e_k] - e_k $ (correct)	(five others)
24T201	#5	3.1×10^{-202}	0.18–0.50
24T203	#3	1.1×10^{-305}	0.18–0.50
24T202	#4	9.8×10^{-296}	0.18–0.50

Table 3: Detection of the Galois-stable synthematic total by rationality. The separation is unambiguous, and the stable total differs from case to case.

source	sextic group	quintic (reduced)	d_K
24T201	A_5	$x^5 - 295x^3 - 2065x^2 - 1475x + 295$	$5^6 \cdot 59^4$
24T203	A_5	$x^5 - 10x^3 - 2x^2 + 19x + 6$	$2^6 \cdot 887^2$
24T202	$\text{PGL}(2, 5)$	$x^5 - 5x^3 + 4x - 1$	38569

Table 4: Quintics produced by (2). All three sextics are totally real. The last row is a negative control: the invariant is S_5 -equivariant and transports whichever group is present.

Remark 7.1 (Two failure modes). An early implementation produced $x^5 - 540x^4 + 108702x^3 - 9977246x^2 + 406832074x - 5881189625$ with $\text{polgalois} = S_5$ for the 24T201 sextic. Both the polynomial and the group were artefacts. First, the total was chosen a priori rather than detected, and was one of the five wrong ones. Second, the code rounded the resulting non-rational coefficients to integers; only e_1 and e_5 happened to be rational, and the middle three were altered from 105725, -9341500 , 381588250 . The S_5 output is moreover self-refuting: the five values (2) lie in a splitting field of degree 60, so no quintic obtained this way can have a splitting field of order 120. Any implementation should therefore assert that the maximal deviation from integrality is below a fixed threshold before rounding.

8 Consequences for icosahedral ramification at 1951

Suppose ρ is even icosahedral, unramified outside a prime $p \nmid 60$. By Lemma 3.4 the quintic resolvent is totally real, and by Corollary 3.3 its discriminant is p^2 or p^4 , the latter requiring $p \equiv \pm 1 \pmod{5}$.

For $p = 1951$ we have $1951 \equiv 1 \pmod{5}$, so both are a priori possible. Theorem 1.1 eliminates $d_K = 1951^2$. Hence:

Corollary 8.1. *If an even icosahedral representation unramified outside 1951 exists, then inertia at 1951 has order 5, the quintic resolvent has $d_K = 1951^4 = 14\,489\,485\,046\,401$, and the Artin conductor of the two-dimensional representation is 1951^2 . In the degree 24 field the discriminant exponent at 1951 is 16 or 20, not 18.*

A direct Hunter search at $d_K = 1951^4$ is infeasible: $B \approx 1846$, so $|e_5| \leq 2.6 \times 10^6$ and the box exceeds 10^{18} points per cell. The practical route is to locate a field of higher degree with the required ramification and descend to the quintic by Section 7.

For $p = 15733$ we have $15733 \equiv 3 \pmod{5}$, so only $d_K = p^2$ is possible, and Theorem 1.2 exhibits the field. This is what gives Theorem 1.1 its force: the same framework yields a positive answer where one is expected and a negative answer at 1951, so the negative result is not an artefact of an over-restrictive setup.

Remark 8.2. We caution that Theorem 1.1 excludes only fields ramified *solely* at 1951. Table 2 shows that A_5 quintics with $1951^2 \mid d_K$ are not rare; eleven occur inside the box. What does not occur is one with no other ramification.

9 Reproducibility

The search is implemented in Python with NumPy and SymPy; verification uses PARI/GP 2.15.4. Cells (a_1, p_2) are independent and are distributed across processes, with completed cells journalled so that interrupted runs resume. The program emits a GP script that computes d_K for every survivor, so that the distinction of Remark 4.1 cannot be overlooked.

Author contributions

Dula posed the problem, supplied the icosahedral context and the target primes, and executed the large-scale runs. Claude Opus 5 derived the inertia constraints of Section 3, designed and implemented the search and its controls, diagnosed the failure modes of Remark 7.1, and drafted the manuscript. Grok 4.6 and Gemini 3.8 contributed the synthematic formulation of Section 7, the transitive-group candidates, and independent review of the intermediate claims. All computational claims were re-derived by at least two of the authors independently.

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